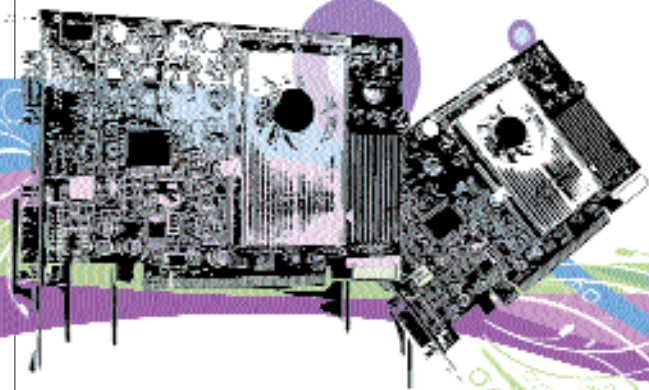


GET REAL!

Graphics Cards are bridging the divide between reality and virtual reality. Usher in the new age by getting one for yourself!



Graphics cards aren't now considered a novelty exclusive to the rich or to gamers. With very tangible real-world benefits, it's time everybody discovered the sheer performance and viewing benefits, not to mention the display flexibility, that a graphics card brings with it. Need we mention that with Vista coming out in a few months, this bit of hardware will become a necessity? Even an upgrade to a very mediocre graphics card from an onboard graphics solution will give you significant performance gains.

MYTHS AND REALITIES

More onboard memory is always better: a general rule of thumb, but not necessarily so. It's the chipset that's important above all else, and then memory. Putting 512 MB of memory on a low-end or mid-range card won't make any difference, since the graphics core will never be able to properly utilise all the memory bandwidth available to it.

Clock speeds: another big myth. With the same chipset, sure, clock speed matters, but across chipsets and cards, it doesn't play a big role. This specification is more limited these days due to heat dissipation. What counts is the number of million / billion pixels and triangles a card can draw per second.

Buying the fastest cards will ensure future proofing: absolutely nothing is future-proof in the world of graphics cards! Even the fastest cards available today will become mediocre if not defunct once DirectX 10 games emerge.

QUESTIONS TO ASK Is a high-end card necessary?

In a word, yes! Because Vista will use 3D graphics for even regular windows, and not just games, a 256 MB DirectX 9 compliant card is the minimum recommended solution for Vista's Ultimate Edition.

The graphics core?

The choice of graphics processor is very important, since it directly affects performance. High-end graphics cores like the G70 and G71 have the following cards based on them: GeForce 7950GX2, 7950GT, 7900GTX, 7900GT, 7800GTX, 7800GT. ATI's R520 and R580 result in the following cards: X1950XTX, X1950XT, X1900XTX, X1900XT, X1800XT. These cores have certain characteristics like compliance to APIs, number of Pixel and Vertex Pipelines, and of course, clock speeds. The performance difference between two different parts based on the same core can be as much as 15 per cent, and with dissimilar cores, more than 100 per cent!

Extra connectivity options?

S-Video and Composite ports, which allow you to output the display to a TV set, are a plus. If you want to record TV, choose a graphics card with an integrated TV-Tuner. DVI connectivity has become essential these days, and a good-quality TFT panel can get a 10 per cent boost in visual quality just by using digital connectivity.

What about AGP cards?

AGP has been largely defunct ever since PCI-Express became mainstream. AGP cards are costlier than PCI-Express cards based on the same chipsets. Buying an AGP card makes no sense today, unless you have an old PC whose video solution just went kaput.



Future Trends

We've all been exposed to the news that DirectX 10 will be the next big catalyst for the gaming industry, and both game developers and graphics card manufacturers are no doubt rubbing their hands in glee. PCI-Express 2.0 is also set to take over from PCI-Express 1.1, though backwards compatibility will be maintained. Supporting 10 GBps—version 1.1 supported 8 GBps—this new standard promises a few optimisations besides the obvious bandwidth advantages. With the kind of raw horsepower that today's graphics cards have, not to mention multi-GPU setups—SLI, Quad SLI and CrossFire—things can only get better. ■



What To Look For

Graphics processor: The core is as important as your CPU is to your system. All those pixel pipelines and the huge pixels per second count is courtesy the GPU (Graphics Processing unit).

DirectX 9.0c (Shader Model 3.0 support): DirectX 9.0c compliance and actual support are different. Make sure of support for Shader Model 3.0. Though DirectX 10 cards are already out, the actual API is still some time away, so it makes sense to wait a bit before investing in these costlier cards.

Dual DVI support: Allows for the connecting of two flat-panel monitors. With the falling price of LCD monitors, you should ensure that your new graphics card features at least one DVI connector.

Memory: 256 MB of video RAM is almost *de facto* these days for a smooth multimedia experience. That said, it makes sense to opt for more memory only if you are planning on getting yourself a much faster graphics card, and if you intend to run a lot of games on it. 512 MB is generally enough!